

DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

FIRMWARE, HOST SOFTWARE & RESULTS

TI C2000 F28069M control models, CAN supervision, acquisition, safeguards, and measured characterization

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.4 | 5 AUGUST 2026

Prepared by	Herder Elektronische Systemen
Document scope	Embedded target behavior, editable Simulink models, CAN command/telemetry, host acquisition, custom ADC integration, and verification
Evidence basis	Target/host Simulink models, custom ADC source, MATLAB scripts, CAN evidence, captured signals, and measured maps
Distribution	Customer-facing; selected source models and controlled engineering evidence included

DOCUMENT PURPOSE

Defines the controlled firmware, host-software, interface, safety, verification, and acceptance requirements. Platform wiring, physical interfaces, schematics, and board ownership are controlled by the companion Platform & Electrical FRD.

Functional Requirements Document Framework

WHY THIS IS AN FRD

This document defines what the embedded and host subsystems shall do, the interfaces and constraints under which they shall operate, and the evidence required for acceptance. Model screenshots and result plots support verification; they do not replace the requirements.

Purpose, Scope, and System Boundary

This FRD governs the C2000 target models, host supervisory models, CAN command and return paths, custom ADC integration, logging, and characterization outputs. Mechanical construction, connector ownership, power wiring, and vendor board details are external interfaces controlled by the companion Platform & Electrical FRD.

Boundary element	Included here	Referenced elsewhere
Target control	Drive/load roles, modes, enable, feedback, ADC acquisition	Board power and physical wiring
Host software	Supervisory commands, acquisition, logging, analysis dispatch	Workstation and licensed-tool provisioning
Verification	CAN tests, model inspection, captured signals, measured characterization	EMC, enclosure, thermal and production qualification

Users and Operational Scenarios

Actor	Primary interaction	Expected outcome
Controls engineer	Modify target/host models within defined interfaces	Behavior remains traceable to requirement IDs
Test engineer	Configure runs, acquire signals, execute analysis	Repeatable evidence products
Customer / reviewer	Inspect requirements, source models and results	Clear demonstrated capability and open limits

1. Embedded and Host Operating Concept

Motor 1 establishes shaft speed while Motor 2 applies an opposing or assisting q-axis current. The C2000 target owns the high-rate field-oriented control loops. Host models provide bounded supervisory references, CAN command packing, acquisition, and test orchestration without becoming the real-time control clock.

- Independent feedback paths are retained for both machines.
- CAN command words enter the existing target reference paths rather than bypassing control logic.
- Custom register-level ADC configuration adds two simultaneous analog-input pairs.
- Host logging retains timestamped signals and metadata for post-run review.

1.1 Operating Modes

Mode	Required behavior
Disabled / safe	Torque-producing references are inhibited.
Drive-only	Motor 1 controls speed; Motor 2 remains at a bounded zero-load condition.
Loaded dynamometer	Motor 1 controls speed while Motor 2 applies bounded q-axis load.
CAN-supervised	Four 16-bit command words are received and mapped to the target command interface.
Acquisition	Selected target signals are recorded without altering real-time loop ownership.

1.2 Requirement Conventions

Mandatory behavior is written as a singular “shall” statement. Each requirement identifies a verification method and release status. “Demonstrated” means observed on retained bench evidence; “Verified” means the release package was inspected and accepted.

2. Firmware and Host Functional Requirements

ID	The system shall...	Verification	Status
FW-001	The embedded target shall execute the fast current and speed loops independently of host refresh, plotting, or logging timing.	Inspection + test	Demonstrated
FW-002	The target shall command one machine as the drive unit and the second as a controllable load or assist unit.	Functional test	Demonstrated
FW-003	The target shall maintain independent encoder feedback paths for both machines.	Inspection + feedback test	Demonstrated
FW-004	The host shall provide bounded speed, load, enable, and motor-selection references.	Model inspection	Demonstrated
FW-005	The host shall support repeatable configuration, capture, and post-run review.	Procedure review	Verified
FW-006	The released source shall include editable Simulink models rather than screenshots alone.	Release inspection	Verified
FW-007	The public release shall exclude generated code, binaries, caches, and proprietary build products.	Release inspection	Verified

2.1 CAN Interface Requirements

ID	The system shall...	Verification	Status
CAN-001	The target shall receive a standard CAN command frame with four 16-bit command words.	Bus test	Demonstrated
CAN-002	The decoded command words shall feed the existing reference and enable paths.	Model inspection	Demonstrated
CAN-003	The target shall provide a return identifier for echo or diagnostic telemetry.	Bus test	Demonstrated
CAN-004	Loss of CAN traffic shall not transfer real-time loop ownership to the host.	Architecture inspection	Verified

3. Custom ADC Requirements

The handwritten F28069M ADC configuration extends the model-based application with two additional simultaneous channel pairs. This source is retained separately from generated MathWorks support code.

ID	The system shall...	Verification	Status
ADC-001	The custom source shall configure SOC12/SOC13 for ADCINA2/ADCINB2 simultaneous sampling.	Source inspection	Verified
ADC-002	The custom source shall configure SOC14/SOC15 for ADCINA1/ADCINB1 simultaneous sampling.	Source inspection	Verified
ADC-003	The added SOC pairs shall use the ePWM1 trigger and a seven-clock acquisition window.	Source inspection	Verified
ADC-004	EOC13 and EOC15 shall route completion events to ADCINT5 and ADCINT9 respectively.	Source inspection	Verified
ADC-005	The public repository shall identify the file as custom register-level integration, not generated firmware.	Document review	Verified

3.1 ADC Acceptance Mapping

SOC pair	Inputs	Trigger	Completion	Evidence
12/13	A2 / B2	ePWM1	EOC13 → ADCINT5	F28069M_ADC_Custom_Config.c
14/15	A1 / B1	ePWM1	EOC15 → ADCINT9	F28069M_ADC_Custom_Config.c

4. Acquisition and Characterization Requirements

This FRD controls the required outputs and acceptance logic. Detailed processing equations and engineering interpretation remain in the implementation notes and scripts, while the retained plots below provide objective verification evidence.

ID	The system shall...	Verification	Status
DAT-001	The workflow shall expose speed, current, voltage, torque-related, and power-related quantities required for review.	Signal inspection	Demonstrated
DAT-002	The analysis shall retain explicit validity masks before map binning.	Script inspection	Verified
DAT-003	Published maps shall distinguish measured bins from unmeasured regions.	Figure inspection	Verified
DAT-004	The release shall retain a sample-count view so sparse operating regions remain visible.	Figure inspection	Verified
DAT-005	Published figures shall be traceable to retained scripts and approved evidence outputs.	Repository inspection	Verified

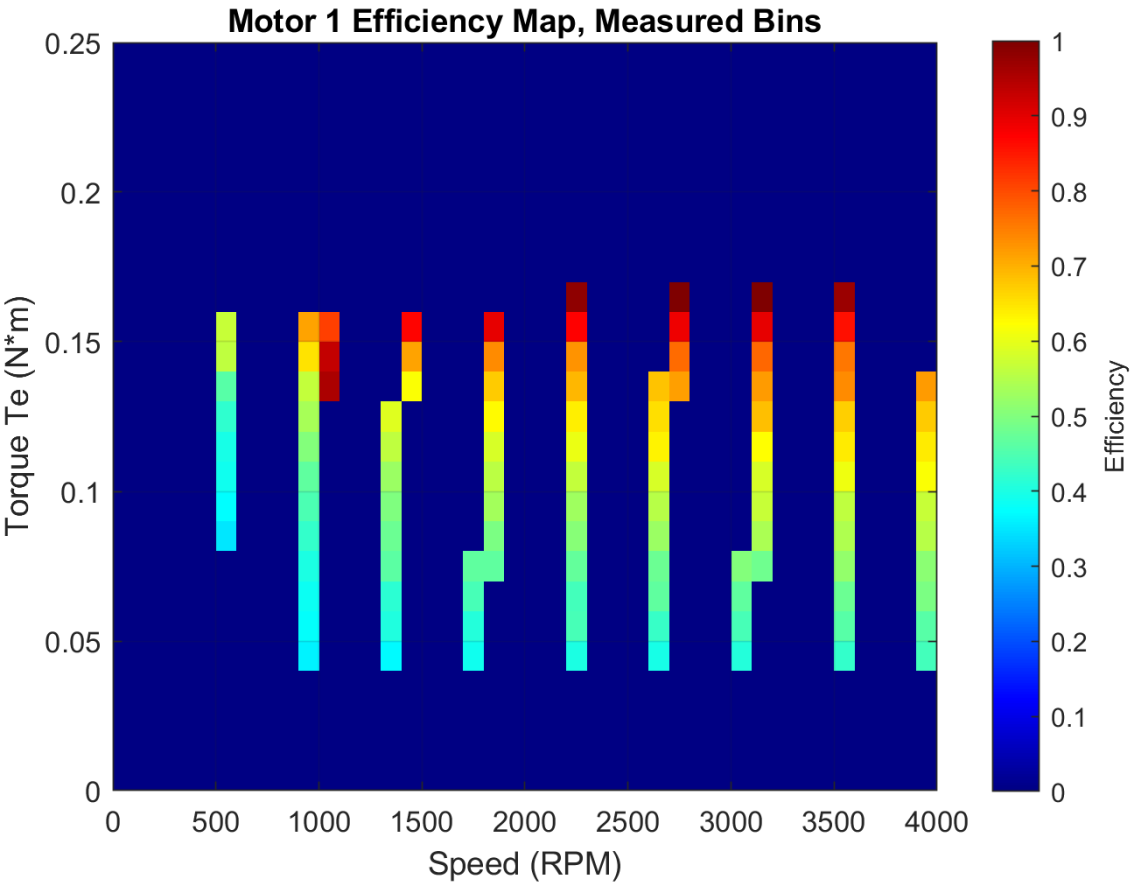


Figure 1. Measured-bin efficiency map retained as verification evidence.

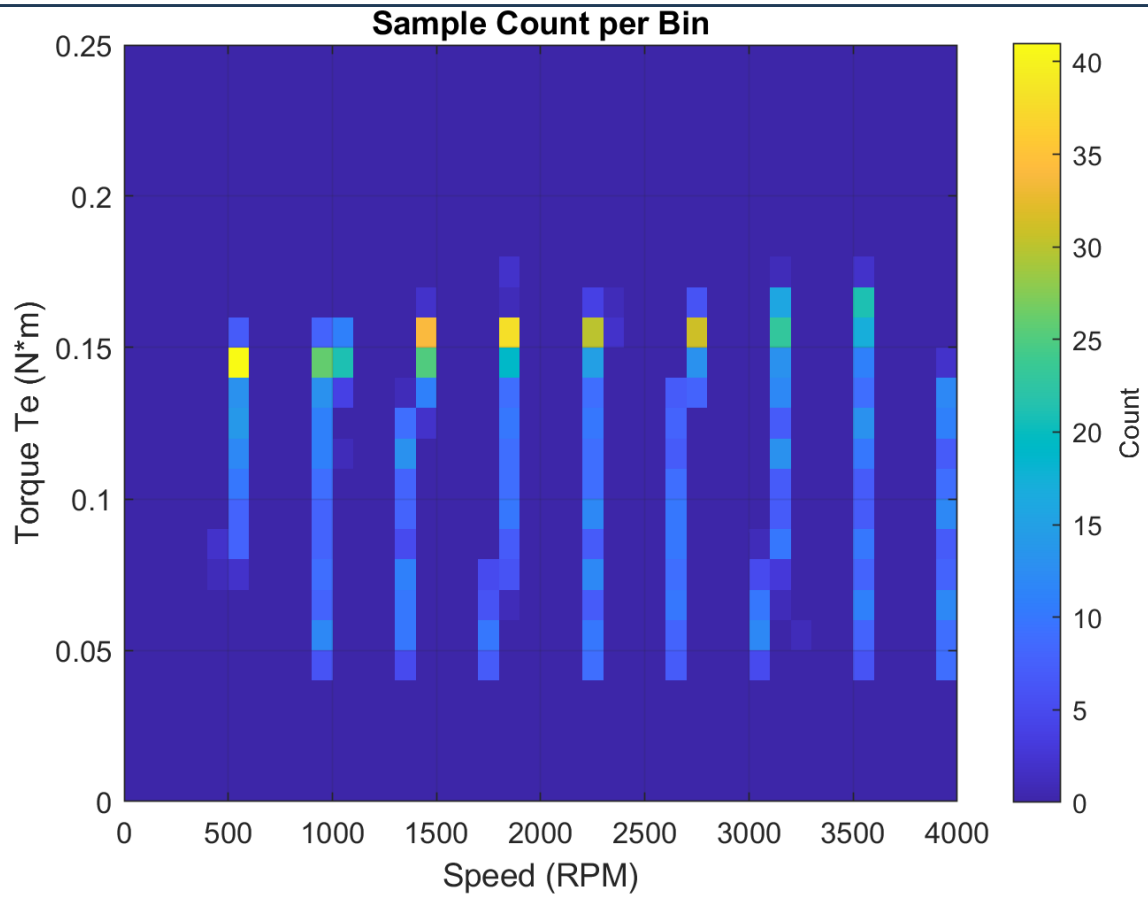


Figure 2. Sample-count map showing evidence density by operating region.

5. Verification and Traceability

Verification uses inspection, analysis, demonstration, and bench test. Acceptance requires a retained evidence path for each mandatory requirement and explicit identification of open qualification items.

Requirement group	Primary method	Retained evidence	Acceptance basis
FW	Model inspection + bench test	Editable .slx models and captured signals	Required control roles and host separation visible
CAN	Model inspection + bus test	Host/target CAN models and diagnostic path	Command receipt and return path demonstrated
ADC	Source inspection	Custom C file	SOC, channel, trigger and interrupt mapping match
DAT	Script + figure inspection	MATLAB source and measured plots	Masks, measured bins and density retained

5.1 Release Limitations

- No calibrated torque traceability is claimed.
- No production functional-safety, EMC, environmental, or endurance qualification is claimed.
- Generated target code and licensed vendor packages are intentionally excluded.
- Efficiency results are bench characterization, not certified machine performance.

5.2 Release Gate

ACCEPTANCE

Release is accepted when every mandatory requirement has a stated method and evidence reference, the editable source models and custom ADC file are present, the public/private boundary is explicit, and both FRDs have been rendered and visually reviewed.